

TOMPO, Russian Federation

WMO: 246710

Lat: 63.950N

Lon: 135.867E

Elev: 402

StdP: 96.59

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-52.1	-50.7	-53.3	0.0	-51.6	-52.0	0.0	-50.5	8.5	-40.1	8.0	-42.8	2.0	350	0.767

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	28.6	17.3	26.3	16.3	24.2	15.4	18.6	25.8	17.4	24.0	16.3	22.5	2.4	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.1	12.0	20.7	14.9	11.1	19.4	13.7	10.3	18.7	54.6	25.9	50.7	24.0	47.2	22.8	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.6	5.4	4.5	-53.2	31.8	1.5	2.3	-54.3	33.5	-55.2	34.8	-56.0	36.1	-57.1	37.8		
(5)			-53.1	20.4	1.6	1.3	-54.2	21.4	-55.1	22.2	-56.0	22.9	-57.1	23.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-12.4	-42.0	-37.8	-25.2	-9.3	4.8	13.4	16.2	12.5	3.5	-12.4	-32.2	-41.5	(6)
(7)		DBStd	22.37	5.94	5.86	7.31	6.54	4.31	3.70	3.55	3.58	4.12	8.18	8.26	6.10	(7)
(8)		HDD10.0	8575	1613	1339	1092	579	166	12	3	16	198	694	1267	1597	(8)
(9)		HDD18.3	11229	1871	1572	1350	829	418	152	83	184	445	952	1517	1856	(9)
(10)		CDD10.0	411	0	0	0	0	6	113	196	93	3	0	0	0	(10)
(11)		CDD18.3	25	0	0	0	0	0	4	18	3	0	0	0	0	(11)
(12)		CDH23.3	632	0	0	0	0	3	161	391	77	0	0	0	0	(12)
(13)		CDH26.7	158	0	0	0	0	0	35	111	12	0	0	0	0	(13)
(14)	Wind	WSAvg	1.9	2.1	2.1	2.0	2.0	1.9	1.8	1.7	1.5	1.7	1.7	2.1	2.0	(14)
(15)	Precipitation	PrecAvg	273	5	5	5	11	25	45	58	53	30	18	12	7	(15)
(16)		PrecMax	412	18	12	30	49	141	82	128	113	78	46	31	18	(16)
(17)		PrecMin	188	1	1	0	1	0	15	2	13	0	3	5	1	(17)
(18)		PrecStd	52	4	2	6	10	25	17	27	25	20	9	6	4	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-22.3	-18.0	-2.1	7.1	23.1	30.0	32.0	28.1	19.5	3.4	-3.0	-20.7	(19)
(20)			MCWB	-22.8	-18.5	-3.9	1.9	11.7	16.4	19.0	17.2	12.0	1.1	-3.3	-21.1	(20)
(21)		2%	DB	-25.8	-21.8	-5.6	5.0	19.3	27.3	29.7	25.4	15.3	1.0	-10.9	-25.6	(21)
(22)			MCWB	-26.0	-22.3	-7.3	0.6	9.8	15.4	17.9	16.4	9.8	-0.8	-11.4	-25.9	(22)
(23)		5%	DB	-29.4	-25.7	-9.3	3.4	16.1	25.1	27.6	23.1	12.6	-0.5	-16.5	-29.3	(23)
(24)			MCWB	-29.5	-26.0	-10.6	-0.3	8.1	14.4	17.4	15.5	8.3	-2.0	-16.8	-29.5	(24)
(25)		10%	DB	-33.3	-28.9	-12.8	1.4	13.6	23.0	25.4	20.7	10.4	-2.0	-20.7	-33.1	(25)
(26)			MCWB	-33.4	-29.1	-13.8	-1.4	6.9	13.7	16.5	14.2	7.4	-3.2	-21.0	-33.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-22.5	-18.4	-3.9	2.5	12.3	18.0	20.6	19.0	12.5	2.1	-3.3	-21.1	(27)
(28)			MCDWB	-22.2	-17.9	-1.9	6.3	21.7	24.9	28.0	25.3	17.5	2.8	-3.2	-20.7	(28)
(29)		2%	WB	-26.0	-22.2	-7.4	1.0	10.2	16.3	19.3	17.2	10.7	-0.3	-11.4	-25.9	(29)
(30)			MCDWB	-25.7	-21.9	-5.5	4.4	17.8	24.9	26.9	23.4	14.0	0.7	-10.9	-25.6	(30)
(31)		5%	WB	-29.5	-25.8	-10.7	0.0	8.7	15.4	18.3	16.0	9.4	-2.0	-16.8	-29.4	(31)
(32)			MCDWB	-29.3	-25.6	-9.4	3.0	15.2	23.3	25.4	21.3	11.9	-0.8	-16.4	-29.3	(32)
(33)		10%	WB	-33.4	-29.0	-13.8	-1.3	7.4	14.5	17.3	14.8	7.6	-3.3	-20.9	-33.2	(33)
(34)			MCDWB	-33.3	-28.9	-12.8	1.5	12.8	21.6	23.8	19.8	10.0	-2.0	-20.6	-33.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.2	10.2	15.9	15.2	11.9	13.6	13.4	13.3	9.4	8.5	7.3	5.5	(35)
(36)			MCDBR	10.9	12.3	16.3	13.6	17.9	19.0	18.9	18.2	14.7	7.8	11.1	10.7	(36)
(37)			MCWBR	10.9	11.9	14.5	9.8	9.7	8.6	8.6	9.6	9.1	6.2	10.7	10.4	(37)
(38)		5% WB	MCDBR	10.9	12.3	16.2	12.1	15.7	16.8	16.0	15.5	12.2	7.2	11.1	10.6	(38)
(39)			MCWBR	10.8	12.0	14.4	8.9	8.8	8.4	8.2	8.8	8.1	6.0	10.7	10.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.179	0.243	0.297	0.364	0.416	0.400	0.493	0.426	0.329	0.261	0.186	0.168	(40)	
(41)		taud	1.927	2.083	2.033	1.914	2.018	2.286	1.993	2.165	2.401	2.207	1.919	1.895	(41)	
(42)		Ebn at Noon	420	678	772	788	783	815	708	730	758	648	380	186	(42)	
(43)		Edh at Noon	28	63	103	148	147	116	151	116	74	55	27	12	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.21	1.10	2.97	4.99	5.35	5.43	5.05	3.94	2.21	1.16	0.37	0.08	(44)	
(45)		RadStd	0.02	0.06	0.14	0.20	0.32	0.47	0.41	0.32	0.27	0.11	0.03	0.01	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page